

# Sylvain Brunet

Programmer specialized in C++ and game engines

Quebec, QC, Canada | (+1) 263 384-8375 | Post-graduation work permit  
[sylvain.brunet37@gmail.com](mailto:sylvain.brunet37@gmail.com) | <https://www.linkedin.com/in/sylvainbrunet37/>  
<https://github.com/SylvainBRUNET37> | <https://sylvainbrunet37.github.io/Portfolio/>

## PROFESSIONAL EXPERIENCE

### 3D Programmer (Anvil Pipeline) - Intern

Ubisoft Quebec | Quebec, QC, Canada | May 2026 to October 2026

- Adding features, debugging, profiling, and optimizing CPU/GPU performance on console in a proprietary engine (Anvil) used by multiple AAA games (Assassin's Creed titles, Ghost Recon, unannounced projects)
- Real-time 3D rendering in C++ and HLSL for VFX, burning, sky, and clouds
- Applied mathematics, PBR, volumetric rendering, raymarching
- Working on systems that allow artists to manipulate rendering and mood/atmosphere
- Agile methodology, code reviews, and use of Perforce and Jira
- Collaboration with international and multicultural teams

### Software Programmer - Intern

Agileo Automation | Poitiers, France | April 2025 to August 2025

- Development of C# gateways integrated into an industrial framework for communication with semiconductor manufacturing equipment
- Integration of unit tests, use of gRPC and OPC-UA for real-time, interoperable client-server applications (multi-language and cross-platform)
- Code reviews, agile methodology, use of Git and Azure DevOps

### Software Programmer - Intern

University of Tours | Tours, France | May 2022 to July 2022

- Requirements analysis, database design, and development of a C++ tool to store and process climatological data

## EDUCATION

### Master's degree in Computer Science for Video Game Development (dual degree)

University of Sherbrooke | Longueuil, QC, Canada | 2025 to 2026

### Engineering Degree in Computer Science (equivalent to a Master's degree)

Polytech Tours | Tours, Centre-Val de Loire, France | 2023 to 2026

## SKILLS

- Languages: C++, HLSL, C# (.NET), C
- Profiling & Debugging: Profilers (PIX, Razor, VTune, proprietary profiler), RenderDoc
- Engine & Platforms: Unreal Engine, console SDKs (Sony/Microsoft), Windows, Linux
- Tools: Git, Perforce, Visual Studio, JetBrains IDEs, Jira, AI (Claude, Cline)

## LANGUAGES

- French: Native
- English: Professional / Technical

## PROJECTS

### **Mon Ami: 3D Multiplayer Horror Game built with Unreal Engine 5 in C++**

*January 2026 to May 2026*

- Developed in a team of 5 programmers and 1 composer
- Client/server architecture with dedicated server
- Data-driven gameplay systems and procedural generation
- Creation of a perception asymmetry system, altering rendering, physics, audio, gameplay, and AI depending on the player
- Spatialized audio (Wwise), debug tools, packaging, backend and Steam integration

### **Fish Engine & Need4Fish: 3D Game Engine & Underwater Game**

*September 2025 to December 2025* | <https://github.com/SylvainBRUNET37/FishEngine>

- 3D game engine built in C++ by a team of 4, using DirectX and Jolt Physics, with a data-oriented (ECS) and data-driven architecture
- Real-time rendering and HLSL shaders, profiling and optimization, resource management, asset pipeline, gameplay systems, and physics engine integration

### **Development of AI algorithms for a programming challenge**

*September 2025 to December 2025* | <https://github.com/Nell15/AI-Bootcamp>

- C++, AI, problem solving, pathfinding, cooperation and interactions between agents, decision-making architectures, state machines, utility-based evaluation (utility AI)

### **Other Projects**

- CHIP-8 emulator in C ([Github](#)), genetic algorithm in C++ ([Github](#))

## INTERESTS

- Video games: particularly online games, Souls-like games, and strategy games
- Reading: Clean Code, Design Patterns, Game Engine Architecture, etc.
- Conferences and events: CppCon, GDC, MIGS, etc.